INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
Mid-Spring Semester Examination 2022-23
Date of Examination: 17.02.23 Session: AN Duration: 2 hrs Full Marks: 40
Subject No.: AI60004 Subject: Big Data processing
Dept/Center/School: Centre of Excellence in Artificial Intelligence
Specific charts, graph paper, log book required: NO,

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1. Answer the questions below. **5x2**
   a. What is the role of jobtracker in Hadoop?
   b. A datanode has 50 data blocks. What is the maximum number of map tasks map-reduce can run?
   c. A map-reduce job is running 10 map tasks and 5 reduce tasks. Three nodes running three reduce tasks die immediately after finishing the task. How many reduce tasks needs to be redone?
   d. The map phase of map-reduce job produces 50 distinct keys. If 100 reducers are run, what is the maximum number of files the map-reduce job is going to produce?
   e. How does Hadoop determine a datanode failure?

3+5+2

2. a) A map-reduce (MR) cluster has 1000 datanodes and it stores a 1 GB data with block of size 1 MB each. A map-reduce job on the data recognized that only 50 blocks are being used repeatedly. What is the amount of cluster DRAM being used by the MR job? Mention amount of memory usage separately for the datanodes and the master node.

   b) A Hadoop cluster has 1000 datanodes in a single rack. The max lifetime of a server is 100 days and within this period a server can fail on a particular day with uniform probability 1/100. The datanodes store a data D that is in 1000 blocks and with replication level 2. A MR job is being run on this cluster on the very first day after installation. What is the expected number of map task rescheduling (due to node failure) that will happen on the first day? Assume one map task for each data block.

   c) What is the main problem with storing too many small files in Hadoop cluster?

3+3+4

3. a) A company has two Hadoop datacentre (say d1 and d2) each having two racks (r1 and r2). Each rack has 1000 datanodes. We use the notation i/rj/dk to denote the i-th datanode of the j-the rack of k-th datacentre. Compute the network distance between the following nodes:
   i) 1/r1/d1 and 2/r1/d1
   ii) 5/r1/d1 and 10/r2/d2
   iii) 1/r2/d2 and 2/r2/d1

   b) The map task of a map-reduce job produces 100 key-value pairs with 10 distinct keys. Assuming that load-balancing is ensured, what is the maximum number of record that a reduce task has to process in the worst case?

   c) What is replica pipeline in Hadoop? How it is created (explain with an example).

4+3+3

4. a) The reduce phase of a map-reduce job requires access to a read only shared variable of 100 MB. Each data node of the Hadoop cluster has 32 GB DRAM and the block size it operates on is 2 GB. The reduce phase needs the shared data for each key-value aggregation. As a programmer suggest a strategy to minimize data communication during the reduce phase. Also mention the amount of additional cluster DRAM your approach is going to consume.

   b) Write down the input and output types for map, reduce and combiner in map-reduce separately.

   c) What is the role of partitioner in map-reduce? Mention two factors that guides the design of partitioner.

You can use the Distributed Cache provided by Hadoop or you can use Apache Spark's Broadcast variables
Additional cluster DRAM = 100 MB * N nodes. But given each Datanode has 32 GB DRAM so this memory overhead is negligible.